G7-D and G8 Status Report - Frozen Version

September 2, 2026

1 I. Resultados Demostrados $\boxed{\textbf{PROVEN}}$

1.1 G7-D - $\boxed{\textbf{PROVEN}}$

Identidades $|\nabla n|^2 = \frac{1}{2}s^2 + \frac{1}{2}t^2 + |b|^2 + |Q|^2$, $(Q^2)^\circ = 0$, $\langle [W, Q], Q\rangle = 0$, gap L^2 e L^1.

1.2 G8-A - $\boxed{\textbf{PROVEN}}$

$\|\Delta n\|_2 \leq \sqrt{(C_E + C_0)/\lambda}$, $C_\lambda = O(\lambda^{-1/2})$.

2 II. Resultados Espectrales Demostrados $\boxed{\textbf{PROVEN}}$

G8-B.1: $|\widehat{\operatorname{inc} M}(k)| = |k|^2 |P_k \widehat{M}(k) P_k|$, $P_k = I - \hat{k} \otimes \hat{k}$, $\|\mathcal{P}M\|_{\dot{H}^1} \leq \|\operatorname{inc} M\|_{\dot{H}^{-1}}$.

3 III. Evidencia Computacional $\boxed{\textbf{COMPUTATIONAL EVIDENCE}}$

$C_\lambda = 104$ para $\lambda = 10^{-3}$ como constante obtenida por implementacion. H_{L^2} : $0.296 \to 0.206$ en 200 iteraciones como evidencia exploratoria.

4 IV. Conjetura $\boxed{\textbf{CONJECTURE}}$

$\|\nabla M\|_2 \lesssim C_\lambda^{3/4} \|\operatorname{inc} M\|_2^{1/4}$ (Conj-G8) is numerically suggested and not claimed as theorem. Target: $\|\nabla M\|_2^4 \lesssim \|M\|_{H^2}^3 \|\operatorname{inc} M\|_2$ (G8.14). A sufficient quartic inequality of form G8.14 would imply Conj-G8 after inserting available H^2 bound.